through, and the state of the contract is updated (e.g., the number is stored). If the transaction doesn't have enough gas, it fails and is reverted. This is why optimising gas usage is a key part of Solidity development.

How to set up your first Solidity project

Setting up your first Solidity project involves a few key steps — from creating the development environment to writing and deploying a simple smart contract. Here is a step-by-step guide on how to do this by creating a dummy project using Solidity and Truffle (a popular framework for Ethereum-based projects).

Install Node.js: First, make sure you have Node.js installed. To check if Node.js is installed, run this command in your terminal:

```
node -v
```

Install Truffle framework: Truffle is a popular development framework for Ethereum. It simplifies the process of developing smart contracts and interacting with the blockchain.

Install Truffle globally using npm (Node Package Manager):

```
npm install -g truffle
```

Verify the installation:

```
truffle version
```

Initialise your project: Now, create a new directory for your project, and then initialise it:

```
mkdir MyFirstSolidityProject
cd MyFirstSolidityProject
truffle init
```

This creates the basic project structure with folders like *contracts*, *migrations*, and *test.*

Set up Ganache: Install Ganache for a local Ethereum blockchain. You can either download the GUI version from Ganache or use Ganache CLI with:

```
npm install -g ganache-cli
```

Now run Ganache CLI:

```
ganache-cli
```

This will give you a local blockchain running on *http://127.0.0.1:8545.*

Write your smart contract: In the *contracts* folder, create a new Solidity file called *MyFirstContract.sol:*

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract MyFirstContract {
string public message;
// Constructor to initialize the message
constructor() {
message = "Hello, World!";
}
// Function to set a new message
function setMessage(string memory newMessage) public {
message = newMessage;
}
}
```

Create a migration script: In the *migrations* folder, create a new migration file named *2_deploy_contracts.js* to deploy the contract:

```
const MyFirstContract = artifacts.require("MyFirstContract");
module.exports = function (deployer) {
deployer.deploy(MyFirstContract);
};
```

Configure Truffle to use Ganache: In the *truffle-config.js file*, configure the development network to use Ganache:

```
Module.exports = {
networks: {
development: {
host: "127.0.0.1", // Localhost
port: 8545, // Ganache default port
Network_id: "*", // Match any network id

},
},
compilers: {
solc: {
version: "0.8.0", // Solidity version
},
},
};
```

Compile your contract: Now compile your smart contract to generate the necessary artifacts:

```
truffle compile
```

Deploy to Ganache: Deploy the contract to your local Ganache blockchain with:

```
truffle migrate --network development
```